

MGT402 Short notes by Kamran

Lec # 23

PROCESS COSTING SYSTEM

(Opening balance of work in process)

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Two methods of cost allocation

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- (1) The weighted average (or averaging) method
- (2) The FIFO method.

Weighted average method

In the weighted average method opening stock values are added to current costs

FIFO method

- It is more complicated to operate
- In process costing, it seems unrealistic to relate costs for the previous period to the current Period of activities

Choosing the valuation method in examinations

In order to use the weighted average or FIFO methods to account for opening work-in-process different information is needed, as follows:

For weighted average An analysis of the opening work-in-process value

Into cost elements (i.e. materials, labor)

For FIFO The degree of completion of the opening work in process for each cost element.

Lec # 25

COSTING/VALUATION OF JOINT AND BY PRODUCTS

Basis of Cost Allocation (For by products)

- (1) Physical Quantity Ratio
- (2) Selling Price Ratio
- (3) Hypothetical Market Value Ratio

Classification of by product

By product can be classified into two categories:

1. Requiring no further process
2. Requiring further processing

Accounting for By Products

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- (1) Income Approach
- (2) Costing Approach

Lec#27

MARGINAL AND ABSORPTION COSTING

(Product costing systems)

Cost Elements

Direct Material, Direct Material
(Variable Cost)

Factory Overhead Cost
(Variable & Fixed Cost)

Marginal Costing

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The cost of a cost unit is presented as the total of direct materials, direct labor, direct expenses and variable overheads (but not fixed overheads)

Marginal cost is the cost the variable cost that changes with the production of each next unit.

(A key concept in marginal costing is that of contribution margin)

Absorption and marginal costing

In **absorption costing**, fixed manufacturing overheads are absorbed into cost units. Thus stock is valued at absorption cost and fixed manufacturing overheads are charged in the profit and loss account of the period in which the units are sold.

In **marginal costing**, fixed manufacturing overheads are not absorbed into cost units, Stock is valued at marginal (or variable) cost and fixed manufacturing overheads are treated as period costs and are charged in the profit and loss account of the period in which the overheads are incurred.

(Under absorption costing stock will include variable and fixed overheads whereas under marginal costing stock will only include variable overheads.)

Contribution Margin

Contribution margin = Sales - variable costs of sales

Contribution margin is short for "contribution to fixed costs and profits"

Marginal costing: Profit calculation

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Sales

Rs X

Less: variable costs	(X)

Contribution margin	X
Less: fixed costs	(X)

Profit	X

In short contribution margin less fixed cost is called profit

Absorption costing: profit calculation

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In absorption costing this is effectively calculated in one stage as the cost of sales already includes fixed costs

Sales	X
Less: absorption cost	(X)

Profit	X

Marginal or absorption costing can be useful for internal management reporting

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If stock levels are rising from opening to closing balance

Absorption Costing profit > Margin Costing profit

(More profit)

(Less profit)

If stock levels are falling from opening to closing balance

Absorption Costing profit < Margin Costing profit

(Less profit)

(More profit)

(Fixed costs carried forward are charged in this period, under absorption costing)

If stock levels are the same

Absorption Costing profit = Margin Costing profit

(same profit)

(Same profit)

Reconciliation formula to learn

Rs.

Profit as per absorption costing system xxx

Add Opening stock @ fixed FOH rate at opening date xxx

Less Closing stock @ fixed FOH rate at closing date xxx

Profit as per marginal costing system

xxx

(The only difference between using absorption costing and marginal costing as the basis of stock valuation is the treatment of fixed production costs.)

Arguments against absorption costing

In absorption costing the fixed costs do not change as a result of a change in the level of activity. Therefore such costs cannot be related to production and should not be included in the stock valuation

Lec# 29

COST – VOLUME – PROFIT ANALYSIS

(Contribution Margin Approach)

___CVP stands for **COST – VOLUME – PROFIT**

CVP analysis may also be used to predict profit levels at different volumes of activity based upon the assumption that costs and revenues exhibit a linear relationship with the level of activity.

Cost-volume-profit analysis determines how costs and profit react to a change in the volume or level of activity, so that management can decide the 'best' activity level.

Following are the assumptions which are used in CVP analysis.

1. Variable costs and selling price (and hence contribution) per unit are assumed to be unaffected by a change in activity level.

2. Fixed costs, whilst not affected in total by a change in the activity level, will change per unit as the activity level changes and there are more (or less) units over which to "share out" the fixed costs if fixed costs per unit change with the activity level, then profit per unit must also change.

CVP is a relationship of four variables

Sales	-----à	Volume
Variable cost	-----à	Cost
Fixed cost	-----à	Cost
Net income	-----à	Profit

Two approaches of CVP analysis

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- (1) Contribution margin approach
- (2) Break even analysis approach

Contribution Margin Approach & CVP Analysis

Contribution margin contributes to meet the fixed cost. Once the fixed cost has been met the incremental contribution margin is the profit

(Income Statement as per the marginal costing system is used as a Standard format of Income Statement to analyze the Cost-Volume-Profit relationship.)

Sales	xxx
Variable Cost	(xxx)

Contribution Margin	xxx
Fixed Cost	(xxx)

Profit	xxx

- (1) Physically increase in volume causes an increase in contribution margin and if there is not increase in the fixed cost because of such change, the incremental contribution margin is added in the final profits
- (2) Increase in sales price per unit causes an increase in the contribution margin, as there is not change in the volume the fixed will remain unchanged. So the incremental change is contribution margin is included into the profit.

- (3) Decrease in sales price per unit causes a decrease in the contribution margin, as there is not change in the volume the fixed will remain unchanged. So the change in contribution margin is subtracted from the profits, which result into a loss

(Normally a decrease in sales price should case an increase in the sales volume).

- (4) Decrease in sales price per unit causes a decrease in the contribution margin, as well as an increase in volume is causing an increase in the profit, this results in an increase in profit



1. At zero contribution margin the loss will be equal to the fixed cost

2. Increase in variable cost reduces the contribution margin

3. Sales – Variable cost = Contribution margin

4. Contribution margin + Variable cost = Sales

5. Contribution margin – Fixed cost = Profit

6. Profit + Fixed cost = Contribution margin

7. Sales - Variable cost = Fixed cost + Profit

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COST – VOLUME – PROFIT ANALYSIS

(Break-even Approach)

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Break-even

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Break-even is the point where sales revenue equals total cost.

In case neither a profit nor a loss

Profit (or loss) is the difference between contribution margin and fixed costs.

Thus the break-even point occurs where contribution margin equals fixed costs



Contribution Margin per unit

Selling price per unit **less** variable costs per unit

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Total contribution

Volume x (Selling price per unit **less** variable costs per unit)

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Target Contribution Margin

Fixed costs + Profit target

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Target Sales in number of units

Target Contribution Margin

Unit contribution

Contribution margin to sales ratio

Contribution to sales ratio(C/S ratio) =Contribution Margin in Rs / Sales in Rs

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Break even sales in Rupees

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% of required amount

Given Amount x _____ = Required Amount

% of given amount

Target CM (fixed cost + target profit)

Break even sales in Rupees = _____

Contribution to sales ratio (CM in Rs / Sale in Rs.)

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Example

If the target contribution margin is equal to Rs. 1,000 then what would be the sales at this point?

Now the above formula will be applied to calculate breakeven sales:

Target CM is the given amount and its % is 25, so the sale which is 100% will be:

100

Rs 1,000 x ---- = Rs 4,000 **break even sale in Rs.**

25

OR

Rs 1,000

----- = Rs. 4,000 **break even sale in Rs**

25

So,

Fixed cost

Break even Sales in Rs. = -----

C/S ratio

Break even sales in units

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Simple formula

Break even sales in Rupees

-----= number of units

Sales price per unit

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Direct formula

Target CM (Fixed costs + Profit target)

CM per unit (Selling price per unit **less** variable costs per unit)

Lec# 31

BREAK EVEN ANALYSIS – MARGIN OF SAFETY

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Margin of Safety (MOS)

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The margin of safety is the difference between budgeted sales volume and break-even sales volume; it indicates the vulnerability of a business to a fall in demand.

Margin of safety=Budgeted sales – Break-even sales

MOS (margin of safety) ratio

MOS (Margin of safety=Budgeted sales)

MOS (margin of safety) ratio = _____ x 100

Budgeted Sales_

OR _____

Budgeted profit

$$\text{Margin of safety ratio} = \frac{\text{Budgeted sales} - \text{Break-even sales}}{\text{Budgeted sales}} \times 100$$

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OR

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$$\text{MOS Ratio} = \frac{\text{Budgeted profit}}{\text{Contribution margin}} \times 100$$

Different ways of calculation of margin of safety



(1) Based on Budgeted sales

$$\text{Margin of safety} = \text{Budgeted sales} - \text{Break-even sales}$$

(2) Using Budget profit

$$\text{Margin of safety} = \frac{\text{Budgeted profit}}{\text{Contribution margin ratio}} \times 100$$

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(3) Using profit and contribution ratio

$$\text{Margin of safety} = \frac{\text{Profit to sales ratio}}{\text{Contribution to sales ratio}} \times 100$$

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Lec#32

BREAKEVEN ANALYSIS – CHARTS AND GRAPHS

The chart or graph is constructed as follows:

- Plot fixed costs, as a straight line parallel to the horizontal axis

- Plot sales revenue and variable costs from the origin

- Total costs represent fixed plus variable costs.

(1) The point at which the sales revenue and total cost lines intersect indicates the breakeven level of output.

(2) By multiplying the sales volume by the unit price at the break-even point the level of revenue needed to break even can be determined.

(3) The chart is normally drawn up to the budgeted sales volume.

(4) The difference between the budgeted sales volume and break-even sales volume is referred to as the margin of safety.



Lec#33

WHAT IS A BUDGET?

Budget

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A budget is a plan expressed in quantitative, usually monetary terms, covering a specific period of time, usually one year.

Two basic classes of budget

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- (1) **Cash budget**
- (2) **Operating budget**

Cash budget

Capital budgets are directed towards proposed expenditures for new projects

And often require special financing.

Operating budget

The operating budgets are directed towards achieving short-term operational goals of the organization, for instance, production or profit goals in a business firm. Operating budgets may be sub-divided into various departmental or functional budgets.

Characteristics of a budget

It is prepared in advance and is derived from the long, term strategy of the organization.

It relates to future period for which objectives or goals have already been laid down.

It is expressed in quantitative form, physical or monetary units, or both.

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Different types of budgets



- (1) Sales. Budget
- (2) Production Budget
- (3) Administrative Expense Budget
- (4) Raw material Budget, etc

All these sectional budgets are afterwards integrated into a master budget- which represents an overall plan of the organization

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A budget helps in following ways

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1. It brings about efficiency and improvement in the working of the organization.
2. way of communicating the plans to various units of the organization. By establishing the divisional, departmental, sectional budgets, exact responsibilities are assigned. It thus minimizes the possibilities of buck-passing if the budget figures are not met.
3. Way or motivating managers to achieve the goals set for the units.

4. It serves as a benchmark for controlling on going. Operations.
5. It helps in developing a team spirit where participation in budgeting is encouraged.
6. It helps in reducing wastage's and losses by revealing them in time for corrective action.
7. It serves as a basis for evaluating the performance of managers.
8. It serves as a means of educating the managers.

Budgetary control



The exercise of control in the organization with the help of budgets is known as budgetary control.

Process of budgetary control

- (i) preparation of various budgets
- (ii) continuous comparison of actual performance with budgetary

Performance

- (iii) Revision of budgets in the light of changed circumstances.

Budget controller

The Chief Executive is finally responsible for the budget programmed, it is better if a large part of the supervisory responsibility is delegated to an official designated as Budget Controller or Budget Director.

Fixation of the Budget Period

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Budget period' means the period for which a budget is prepared and employed. The budget period depends upon the nature of the business and the control techniques.

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Forecast

A forecast is an estimate of the future financial conditions or operating results.

A forecast may be prepared in financial or physical terms for sales, production cost, or other resources required for business. Instead of just one forecast a number of alternative forecasts may be considered with a view to obtaining the most realistic overall plan.

Preparing budgets

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After the forecasts have been finalized the preparation of budgets follows. The budget activity starts with the preparation of the said budget.

Production budget: on the basis of sales budget and the production capacity available

Financial budget (i.e. cash or working capital budget): will be prepared on the basis of sale forecast and production budget.

(All these budgets are combined and coordinated into -a master budget)

Fixed and Flexible Budgets

A fixed budget is based on a fixed volume of activity, it may

lose its effectiveness in planning and controlling if the actual capacity utilization is different from what was planned for any particular unit of time e.g. a month or a quarter.

The flexible budget is more useful for changing levels of activity, as it considers fixed and variable costs separately. Fixed costs, as you are aware, remain unchanged over a certain range of output such costs change when

There is a change in capacity level. The variable costs change in direct proportion to output.

(If flexible budgeting approach is adopted, the budget controller can analyze the variance between actual costs and budgeted costs depending upon the actual level of activity attained during a period of time.)

Objective of Budget

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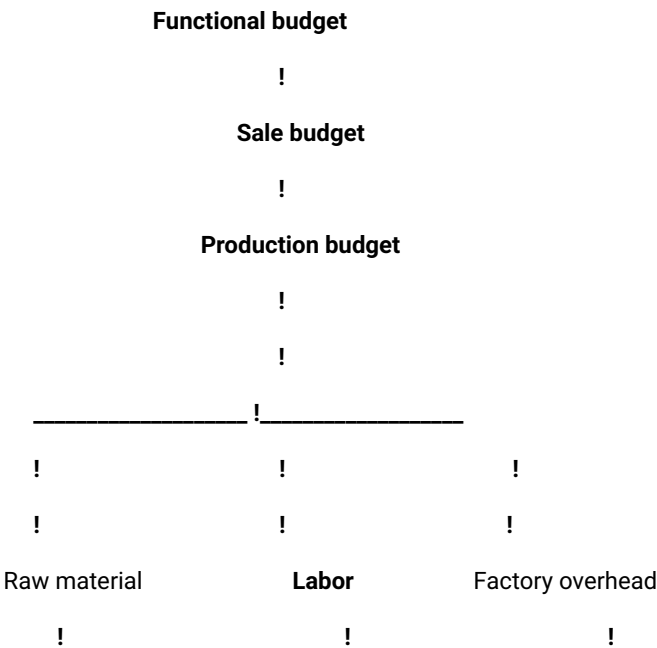
- (1) Profit maximization
- (2) Maximization of sales
- (3) Volume growth
- (4) To compete with the competitors
- (5) Development of new areas of operation.
- (6) Quality of service
- (7) Work-force efficiency.



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Division of Budgets

- (1) **Functional Budget**
- (2) **Master Budget**



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Cost of goods sold

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!	!	!
!	!	!
Selling	general &	financial
Distribution	administrative	charges
Expenses	expenses budget	budget
Budget		

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PRODUCTION & SALES BUDGET

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Budgets can be classified into different categories on the basis of Time, Function, or Flexibility.

Rolling budget

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Budget for a year in advance will always be there. Immediately after a month, or a quarter, passes, as-the case may be, a new budget is prepared for a twelve months. The figures for the month/quarter, which has rolled down, are dropped and the figures for the next month /quarter are added.

Example,

If a budget has been prepared for the year 19X7, after the expiry of the first quarter ending 31st March 19X7, a new budget for the full year ending 31st March, 19X8 will be prepared by dropping the figures for the quarter which has past (i.e. quarter ending 31st March 19X7) and adding the figures for the new quarter ending 31st March 19X8.

Sales budget

Sales Budget generally forms the fundamental basis on which all other budgets are built the budget is based on projected sales to be achieved in a budget period. The Sales Manager is directly responsible for the preparation and execution of this budget.

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Lec#35

PRODUCTION & SALES BUDGET (Contd.)

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Production budget

This budget provides an estimate of the total volume of production distributed product-wise with the scheduling of operations by days, weeks and months and a forecast of the inventory of finished products.

Generally, the production budget is based on the sales -budget.

The production budget is prepared after taking into consideration several factors like:

- (i) Inventory policies.
- (ii) Sales requirements
- (iii) Production stability
- (iv) Plant capacity
- (v) Availability of materials and labor
- (vi) Time taken in production process, etc.

Lec#36

FLEXIBLE BUDGET (not completed).

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The preparation of a flexible budget results from the development of formulas for each department and for each account within a department or cost center. The formula for each account indicates the fixed amount and/or a variable rate. The fixed amount and variable rate remain constant within prescribed ranges of activity. The variable portion of the formula is a rate expressed in relation to a base such as direct labor hours, direct labor cost or machine hours.

Capacity and volume

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The terms "capacity" and "volume" (or activity) are used in connection with the construction and use of both fixed and flexible budgets.

Capacity is that fixed amount

Volume is the variable factor in business.

Theoretical Capacity

The theoretical capacity of a department is its capacity to produce at full speed without interruptions

Expected Actual Capacity

Expected actual capacity is based on a short-range outlook. The use of expected actual capacity is feasible with firms whose products are of a seasonal nature» and market and style changes allow price adjustments according to competitive conditions and customer demands.

Lec#37

FLEXIBLE BUDGET (Contd.)

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Analysis of Cost Behavior

The success of a flexible budget depends upon careful study and analysis of the relationship of expenses to volume of activity or production and results in classifying expenses as fixed, variable, and semi variable,

Fixed Expenses

A fixed expense remains the same in total as activity increases or decreases.

In the short run, some fixed expenses, some times called programmed fixed expenses, will change because of changes in the volume of activity or for such reasons as changes in the number and salaries of the management groups.

Variable Expenses

A variable expense is expected to increase proportionately with an increase in activity and decrease proportionately with a decrease in activity.

Variable expenses include the cost of supplies, indirect factory labor, receiving, storing, rework, perishable tools, and maintenance of machinery and tools. A measure of activity – such as direct labor hour or dollars.

TYPES OF BUDGET

Cash Budget

Determining the future is a summary of the firm's expected cash inflows and outflows over a particular period of time.

Objective

Cash budget is to enable the firm to meet all its commitments in time
And at the same time prevent accumulation at any lime of unnecessary large cash balances with it:

Format of Cash Budget

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XYZ Ltd
Cash budget
For the month of XXX – XXX

Opening balance

Add Receipts (Anticipated cash

Receipt from all sources)

Less Payments (Anticipated utilization of cash)

Excess / Deficit

Bank barrowing / Overdraft

Closing balance

Lec#39

COMPLEX CASH BUDGET & FLEXIBLE BUDGET

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Flexible budget

The Flexible Budget is designed to change in accordance with the level of activity attained. Such a budget is prepared after considering the fixed and

Variable elements of cost and the changes that may be expected for each item at various levels of Operations



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LEC#40

FLEXIBLE & ZERO BASE BUDGETING

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Controlling ratios

Budget is a part of the planning process.

(1) Activity Ratio

(2) Capacity Ratio

(3) Efficiency Ratio

If the ratio works out to 100 per cent or more, the trend is taken as favorable, if the ratio is less than 100 per cent, the indication is taken

as unfavorable.

Activity Ratio:

Standard hours for actual production

$$\text{Activity Ratio} = \frac{\text{Standard hours for actual production}}{\text{Budgeted hours}} \times 100$$

Capacity Ratio:

Actual hours worked

$$\text{Capacity Ratio} = \frac{\text{Actual hours worked}}{\text{Budgeted hours}} \times 100$$

Efficiency Ratio:

Standard hours for actual production

$$\text{Efficiency Ratio} = \frac{\text{Standard hours for actual production}}{\text{Actual hours worked}} \times 100$$

Performance budgeting

A performance budget presents the operations of an organization in terms of functions, programmes, activities, and projects.

The primary purpose of traditional budget particularly in government administration is to ensure financial control and meet the requirements of legal accountability, that is, to ensure that appropriation.

Objectives of PB

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(1) to coordinate the physical and financial aspects

(2) To improve the budget formulation, review and decision-making at all levels of management

(3) To facilitate better appreciation and review by controlling Authorities (legislature, Board of Trustees or Governors, etc.) as the presentation is more

Purposeful and intelligible

___ (4) To make more effective performance audit possible

(5) To measure progress towards long-term objectives which are envisaged in a development plan

Zero base Budgeting

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Zero base Budgeting technique suggests that an organization should not only make decisions about the proposed new programmes, but should also, from time to time, review the appropriateness of the existing.

The concept of Zero base Budgeting has been accepted for adoption in the departments of the Central Government and some State Governments.



Lec#41

DECISION MAKING IN MANAGEMENT ACCOUNTING

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Costs appropriate to a specific management decision

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The amount by which costs increase and benefits decrease as a direct result of a specific management decision' **Relevant benefits** are 'the amounts by which costs decrease and benefits increase as a direct result of a specific management decision'

Incremental costs

An incremental cost can be defined as a cost which is specifically incurred by following a course of action and which is avoidable if such action is not taken.

Non-incremental costs

These are costs, which will not be affected by the decision at hand. Non-incremental costs are non-relevant costs because they are not related to the decision at hand

The labor cost is non-relevant

Opportunity costs (relevant cost)

An opportunity cost is a level of profit or benefit foregone by the pursuit of a particular course of action.

Sunk cost(non-relevant cost)

A sunk cost is a cost that has already been incurred and cannot be altered by any future decision.

Sunk costs are the opposite of opportunity costs in that they are not incorporated in the decision making process even though they have already been recorded in the books and records of the enterprise.

LEC#42

DECISION MAKING

Q: why companies close down temporarily?

ANS: Companies are often faced with the problem of whether to close down temporarily a part of the plant during periods of low demand.

Arguments against shut-down

(a) If the company continues operation, expenses that would be incurred with the closing down of the plant will be saved; e.g. an increase in factory security.

(b) Continued operation means saving the expenses that will otherwise be incurred if the plant is reopened again at a later stage.

(c) A shut-down for a short period of time will not eliminate all costs. Rent, rates, depreciation and insurance will have to be incurred during the shutdown period.

(d) If the factory is shut down, this will affect not only morale but also its market standing if it cannot meet consumer demand.

The role of fixed costs

If the decrease or increase in the level of activity affects fixed costs then these costs should be considered differential costs.

It is generally accepted that if the plant has excess capacity then new or additional volume may be accepted if the selling price is greater than variable costs. In such a situation, fixed costs are not relevant if they remain fixed at an increased level of output

The role of variable costs

In differential cost studies, if the plant is not operating at practical capacity owing to lack of orders, variable costs usually represent the differential cost whether they are incremental or avoidable. The term refers to those costs that will change. It is often assumed that the variable cost per unit will remain constant regardless of the level of activity.

Lec#43

DECISION MAKING (Contd.)

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Relevant Costs

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Decision making should be based on relevant costs.

- Relevant costs are future costs
- Relevant costs are cash flows
- Relevant costs are incremental costs

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Differential Costs and Opportunity Costs

Relevant costs are also differential costs and opportunity costs.

- Differential cost is the difference in total cost between alternatives.

For example, if decision option A costs Rs. 300 and decision option B costs Rs. 360, the differential costs is Rs. 60.

- An opportunity cost is the value of the benefit sacrificed when one course of action is chosen in preference to an alternative.

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Controllable and Uncontrollable Costs

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Controllable costs are items of expenditure which can be directly influenced by a given manager within a given time span.

As a general rule, committed fixed costs such as those costs arising from the possession of plant, equipment and buildings (giving rise to depreciation and rent) are largely uncontrollable in the short term because they have been committed by longer-term decisions.

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Fixed and Variable Costs

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- Variable costs will be relevant costs.
- Fixed costs are irrelevant to a decision

Attributable Fixed Costs

There might be occasions when a fixed cost is a relevant cost, and you must be aware of the distinction 'specific' or 'directly attributable' fixed costs, and general fixed overheads

Absorbed Overhead and fixed overhead

Absorbed overhead is a national accounting cost and hence should be ignored for decision making purposes.

It is overhead incurred which may be relevant to a decision.

General fixed overheads are those fixed overheads which will be unaffected by decisions to increase or decreased the scale of operations, perhaps because they are an apportioned share of the fixed costs of items which would be completely unaffected by the decision.

General fixed overheads are not relevant in decision making.

Lec#44

DECISION MAKING

CHOICE OF PRODUCT (PRODUCT MIX) DECISIONS

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Make or Buy Decisions and Limiting Factors

In a situation where a company must subcontract work to make up a shortfall in its own production capability, its total costs are minimized if those components/products subcontracted are those with the lowest extra variable cost of buying per unit of limiting factor saved by buying.

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Lec#45

DECISION MAKING (Contd.)

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Shut Down Decisions

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- (1) Whether or not to shut down a factory, department, or product line either because it is making a loss or it is too expensive to run.
- (2) If the decision is to shut down, whether the closure should be permanent or temporary.

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ONE-OFF CONTRACTS

The decision to accept or reject a contract should be made on the basis of whether or not the contract increases contribution and profit.

Other factors to consider in the one-off contract decision.

A. The acceptance of the contract at a lower price may lead other customers to demand lower prices as well.

B. There may be more profitable ways of using the spare capacity.

C. Accepting the contract may lock up capacity that could be used for future full-price business.

D. Fixed costs may, in fact, if the contract is accepted.



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